B2 TP09 Serveur dédie web/ftp :

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Travail à faire :

1) Installer un serveur WEB/FTP sur une machine Ubuntu light.
   a) **serveur web**
   b) **serveur ftp**
   c) **sécurisation du ftp**
   d) **sécurisation https**
   e) **redirection de port http vers https**

Vous devrez **adapter les documentations (versions php, apache...)** des annexes à votre configuration **en vue de l'utilisation dans le cadre de gsb**

2) Procéder à une **phase de tests** afin de vérifier le bon fonctionnement des serveurs

3) Présenter le **schéma réseau** qui a permis de mettre en œuvre votre travail

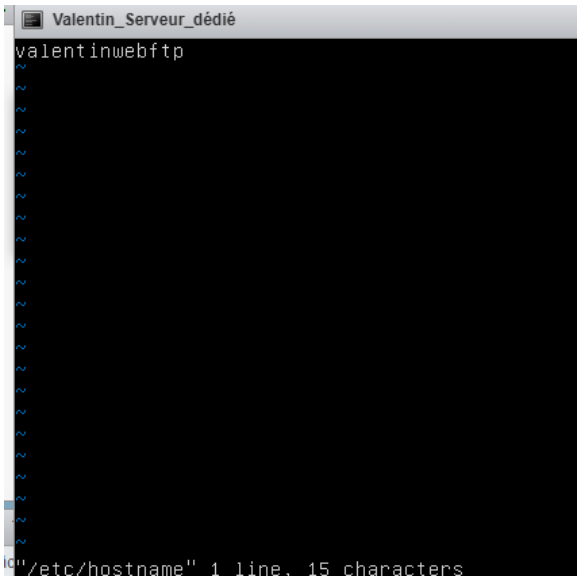
1) a) **serveur web** :

On commence d'abord par rajouter une adresse et changer le nom de l'hôte :

Valentin_Serveur_dédié

```
# interfaces(5) file used by ifup(8) and ifdown(8)
# Include files from /etc/network/interfaces.d:
#source-directory /etc/network/interfaces.d

auto eth0
iface eth0 inet static
     address 172.30.15.175
     netmask 255.255.0.0
     network 172.30.0.0
     broadcast 172.30.255.255
     gateway 172.30.255.254
     dns-nameservers 172.17.63.131 8.8.8.8 8.8.4.4
~
```



(Correction de l'erreur extrajunk at the end : rajouter un « ; »)

On fait les mises à jour :

```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get install ethtool
```

On install ssh pour utiliser putty :

apt install openssh-server -y

On modifie la ligne qui empêche la connection en root :

vi /etc/ssh/sshd_config

Ligne à modifier → PermitRootLogin yes

Et on restart ssh :

service ssh restart

Installations des paquets :

```
root@valentinwebftp:~# apt-get install apache2 php5 libapache2-mod-php5 php5-mys
ql mysql-server -y
```

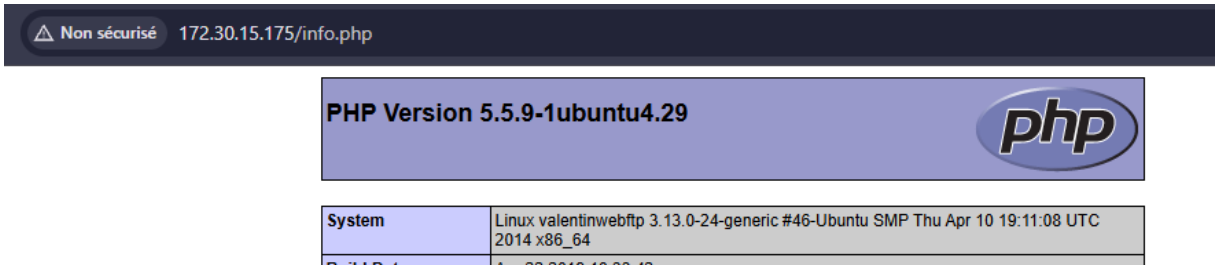
```
root@valentinwebftp:~# apache2 -v
Server version: Apache/2.4.7 (Ubuntu)
Server built:   Apr  3 2019 18:04:25
root@valentinwebftp:~# php -v
PHP 5.5.9-1ubuntu4.29 (cli) (built: Apr 22 2019 18:33:52)
Copyright (c) 1997-2014 The PHP Group
Zend Engine v2.5.0, Copyright (c) 1998-2014 Zend Technologies
    with Zend OPcache v7.0.3, Copyright (c) 1999-2014, by Zend Technologies
root@valentinwebftp:~#
```

Test apache :



Test php :

```
root@valentinwebftp:~# echo '<?php phpinfo(); ?>' > /var/www/html/info.php
root@valentinwebftp:~#
```



b) **serveur FTP** :

Installer Pure-FTPd :

```
root@valentinwebftp:~# apt-get install pure-ftpd pure-ftpd-common -y
```

Créer l'utilisateur et le groupe système :

```
root@valentinwebftp:~# groupadd ftpgroup
root@valentinwebftp:~# useradd -g ftpgroup -d /dev/null -s /bin/false ftpuser
```

Configurer l'authentification :

Désactiver PAM et la résolution DNS :

```
root@valentinwebftp:~# echo no > /etc/pure-ftpd/conf/PAMAuthentication
root@valentinwebftp:~# echo yes > /etc/pure-ftpd/conf/DontResolve
root@valentinwebftp:~# ln -s /etc/pure-ftpd/conf/PureDB /etc/pure-ftpd/auth/50pu
red
root@valentinwebftp:~#
```

Créer le dossier FTP de test et le nouvel utilisateur :

```
root@valentinwebftp:~# mkdir -p /home/ftp/testftp
root@valentinwebftp:~# chown -R ftpuser:ftpgroup /home/ftp/testftp
root@valentinwebftp:~# pure-pw useradd valftp -u ftpuser -g ftpgroup -d /home/ft
p/testftp
Password:
Enter it again:
root@valentinwebftp:~#
```

(mdp 123)

Générer la base et redémarrer :

```
root@valentinwebftp:~# pure-pw mkdb
root@valentinwebftp:~# service pure-ftpd restart
Restarting ftp server: Running: /usr/sbin/pure-ftpd -l puredb:/etc/pure-ftpd/pur
eftpd.pdb -O clf:/var/log/pure-ftpd/transfer.log -E -H -8 UTF-8 -J ALL:!aNULL:!S
SLv3 -u 1000 -B
root@valentinwebftp:~#
```

Vérification :

```
root@valentinwebftp:~# pure-pw list
valftp               /home/ftp/testftp/./
root@valentinwebftp:~# ss -tlnp | grep :21
LISTEN     0      9                            *:21                         *:*
users:(("pure-ftpd",7774,3))
LISTEN     0      9                           :::21                        :::*
users:(("pure-ftpd",7774,4))
root@valentinwebftp:~#
```

c) **sécurisation du FTP** :

Installer OpenSSL et préparer le dossier :

```
root@valentinwebftp:~# apt-get install openssl -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
openssl is already the newest version.
openssl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@valentinwebftp:~# mkdir -p /etc/ssl/private
root@valentinwebftp:~#
```

Générer le certificat :

openssl req -x509 -days 5000 -nodes -newkey rsa:2048 -keyout

/etc/ssl/private/pure-ftpd.pem -out /etc/ssl/private/pure-ftpd.pem

Mettre l'IP de la VM dans common name :

```
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:172.30.15.175
```

Sécuriser le fichier du certificat :

```
root@valentinwebftp:~# chmod 600 /etc/ssl/private/pure-ftpd.pem
root@valentinwebftp:~#
```

Activer TLS :

```
root@valentinwebftp:~# echo 2 > /etc/pure-ftpd/conf/TLS
```

Redémarrer et vérifier :

```
root@valentinwebftp:~# service pure-ftpd restart
Restarting ftp server: Running: /usr/sbin/pure-ftpd -l puredb:/etc/pure-ftpd/pur
eftpd.pdb -O clf:/var/log/pure-ftpd/transfer.log -E -Y 2 -H -8 UTF-8 -J ALL:!aNU
LL:!SSLv3 -u 1000 -B
root@valentinwebftp:~# cat /etc/pure-ftpd/conf/TLS
2
root@valentinwebftp:~# ss -tlnp | grep :21
LISTEN     0      9                          *:21                       *:*
users:(("pure-ftpd",7815,3))
LISTEN     0      9                         :::21                      :::*
users:(("pure-ftpd",7815,4))
root@valentinwebftp:~#
```

d) **sécurisation https** :

Générer le certificat auto-signé :

openssl req -new -x509 -days 365 -nodes -out

/etc/ssl/certs/tpserver.crt -keyout /etc/ssl/private/tpserver.key

```
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:172.30.15.175
Email Address []:
root@valentinwebftp:~#
```

Puis on sécurise la clé :

```
root@valentinwebftp:~# chmod 440 /etc/ssl/private/tpserver.key
```

Activer le module SSL :

```
root@valentinwebftp:~# a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create s
elf-signed certificates.
To activate the new configuration, you need to run:
  service apache2 restart
root@valentinwebftp:~#
```

Créer le VirtualHost HTTPS :

```
root@valentinwebftp:~# vi /etc/apache2/sites-available/tp-ssl.conf
```

```
root@valentinwebftp: ~
<VirtualHost *:443> DocumentRoot /var/www/html ServerName 172.30.15.175 ServerSignature
Off ErrorLog ${APACHE_LOG_DIR}/error_tp.log LogLevel info CustomLog ${APACHE_LOG_DIR}/ac
cess_tp.log combined SSLEngine on SSLCertificateFile /etc/ssl/certs/tpserver.crt SSLCert
ificateKeyFile /etc/ssl/private/tpserver.key </VirtualHost>
~
```

Activer le site et recharger Apache :

```
root@valentinwebftp:~# cat > /etc/apache2/sites-available/tp-ssl.conf << 'EOF'
> <VirtualHost *:443>
>     DocumentRoot /var/www/html
>     ServerName 172.30.15.175
>
>     ServerSignature Off
>     ErrorLog ${APACHE_LOG_DIR}/error_tp.log
>     LogLevel info
>     CustomLog ${APACHE_LOG_DIR}/access_tp.log combined
>
>     SSLEngine on
>     SSLCertificateFile /etc/ssl/certs/tpserver.crt
>     SSLCertificateKeyFile /etc/ssl/private/tpserver.key
> </VirtualHost>
> EOF
```

```
    13  </VirtualHost>
root@valentinwebftp:~# apachectl configtest
AH00557: apache2: apr_sockaddr_info_get() failed for valentinwebftp
AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
 using 127.0.0.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK
```

```
root@valentinwebftp:~# a2ensite tp-ssl
Enabling site tp-ssl.
To activate the new configuration, you need to run:
  service apache2 reload
root@valentinwebftp:~# service apache2 reload
 * Reloading web server apache2
 *
root@valentinwebftp:~# ss -tlnp | grep :443
LISTEN     0      128                :::443                     :::*      users:((
"apache2",7944,6),("apache2",7943,6),("apache2",7942,6),("apache2",7941,6),("apache2",79
40,6),("apache2",7444,6))
root@valentinwebftp:~#
```

Test navigateur :



e) **Redirection de port http vers https** :

Éditer le VirtualHost par défaut en ajoutant la ligne de redirection :

```
root@valentinwebftp:~# sed -i '/<\/VirtualHost>/i\    Redirect permanent / https://172.3
0.15.175/' /etc/apache2/sites-available/000-default.conf
```

```
root@valentinwebftp:~# grep -v '^\s*#' /etc/apache2/sites-available/000-default.conf
<VirtualHost *:80>

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html


        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

    Redirect permanent / https://172.30.15.175/
</VirtualHost>

root@valentinwebftp:~#
```

```
root@valentinwebftp:~# apachectl configtest
AH00557: apache2: apr_sockaddr_info_get() failed for valentinwebftp
AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
 using 127.0.0.1. Set the 'ServerName' directive globally to suppress this message
Syntax OK
root@valentinwebftp:~# service apache2 restart
 * Restarting web server apache2
AH00557: apache2: apr_sockaddr_info_get() failed for valentinwebftp
AH00558: apache2: Could not reliably determine the server's fully qualified domain name,
 using 127.0.0.1. Set the 'ServerName' directive globally to suppress this message
                                                                                 [ OK ]
root@valentinwebftp:~#
```

Tester dans le navigateur :

```
root@valentinwebftp:~# curl -I http://172.30.15.175
HTTP/1.1 301 Moved Permanently
Date: Fri, 06 Nov 2026 10:40:29 GMT
Server: Apache/2.4.7 (Ubuntu)
Location: https://172.30.15.175/
Content-Type: text/html; charset=iso-8859-1
```

La redirection fonctionne correctement.

2) **Phases de tests** :

État des services :

```
root@valentinwebftp:~# service apache2 status
 * apache2 is running
root@valentinwebftp:~# service pure-ftpd status
 * pure-ftpd is running
root@valentinwebftp:~#
```

Ports ouverts :

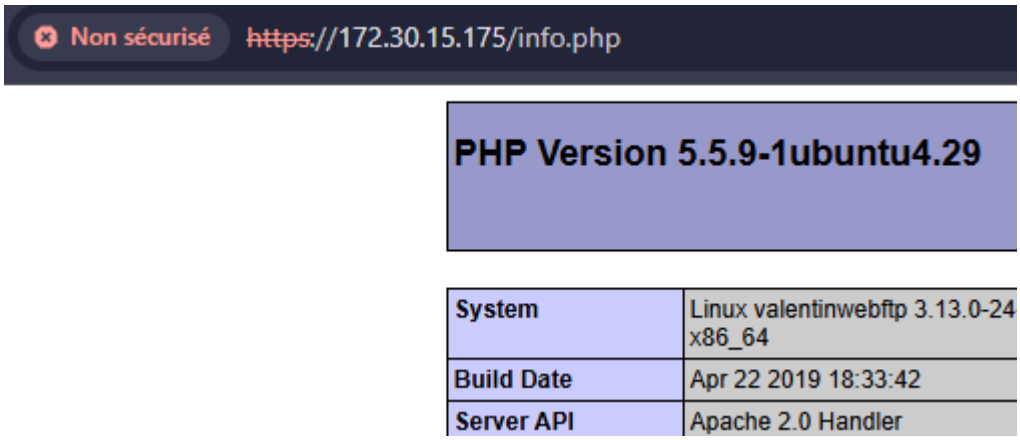
```
root@valentinwebftp:~# ss -tlnp | grep -E ':21|:80|:443'
LISTEN     0      9                          *:21                       *:*      users:((
"pure-ftpd",7815,3))
LISTEN     0      9                         :::21                      :::*      users:((
"pure-ftpd",7815,4))
LISTEN     0      128                       :::443                     :::*      users:((
"apache2",8144,6),("apache2",8077,6),("apache2",8072,6),("apache2",8071,6),("apache2",80
70,6),("apache2",8069,6),("apache2",8068,6),("apache2",8064,6))
LISTEN     0      128                       :::80                      :::*      users:((
"apache2",8144,4),("apache2",8077,4),("apache2",8072,4),("apache2",8071,4),("apache2",80
70,4),("apache2",8069,4),("apache2",8068,4),("apache2",8064,4))
root@valentinwebftp:~#
```

Versions installées :

```
root@valentinwebftp:~# apache2 -v | head -1
Server version: Apache/2.4.7 (Ubuntu)
root@valentinwebftp:~# apache2 -v | head -1
Server version: Apache/2.4.7 (Ubuntu)
root@valentinwebftp:~# pure-ftpd --help | head -1
pure-ftpd v1.0.36 [privsep]
root@valentinwebftp:~#
```

Tests du serveur web :

http://172.30.15.175 redirige bien sur https://172.30.15.175

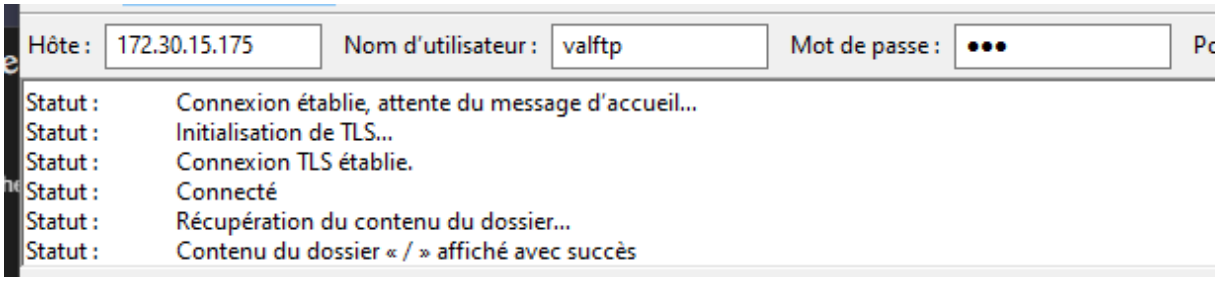


Test curl :

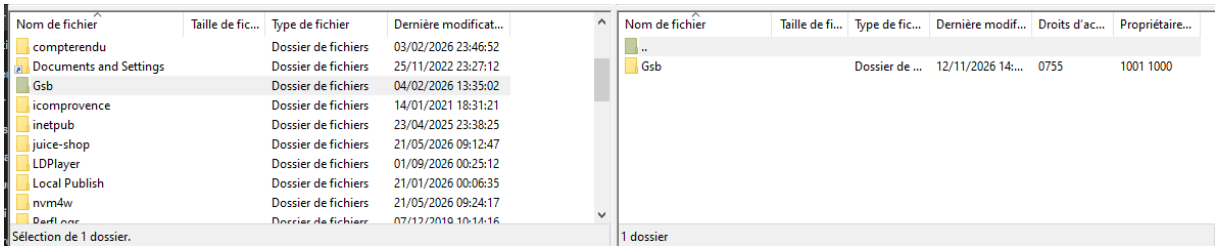
```
root@valentinwebftp:~# curl -I http://172.30.15.175
HTTP/1.1 301 Moved Permanently
Date: Fri, 06 Nov 2026 12:41:45 GMT
Server: Apache/2.4.7 (Ubuntu)
Location: https://172.30.15.175/
Content-Type: text/html; charset=iso-8859-1

root@valentinwebftp:~#
```

Tests du serveur FTP :



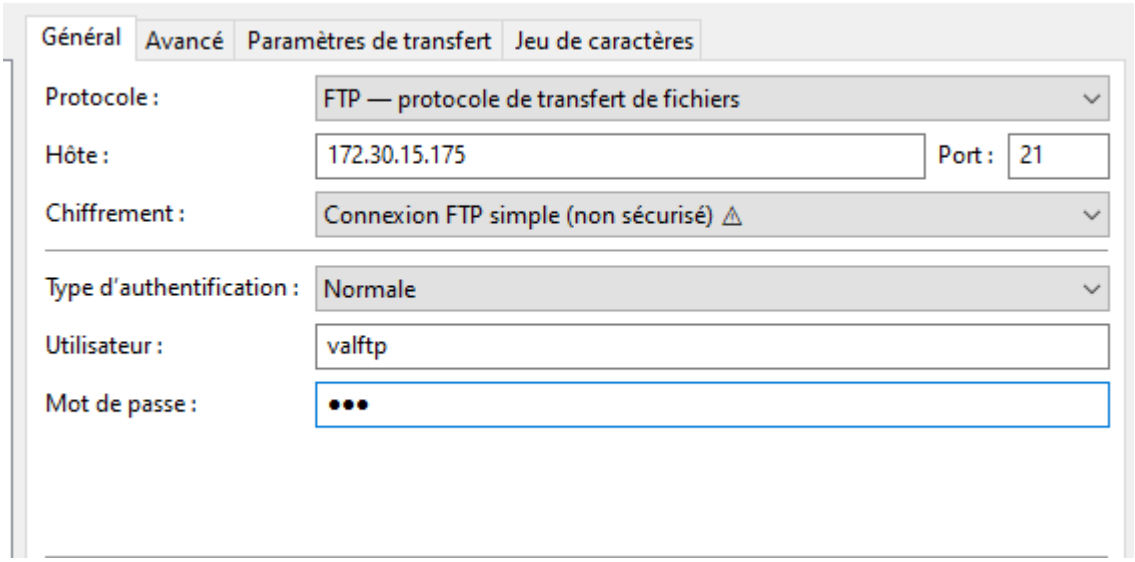
Dépôt d'un dossier :



Vérification de l'ajout :

```
root@valentinwebftp:~# ls -l /home/ftp/testftp
total 4
drwxr-xr-x 6 ftpuser ftpgroup 4096 Nov 12 13:15 Gsb
root@valentinwebftp:~#
```

Test accès non sécurisé :



La connexion est bien refusée comme prévu :

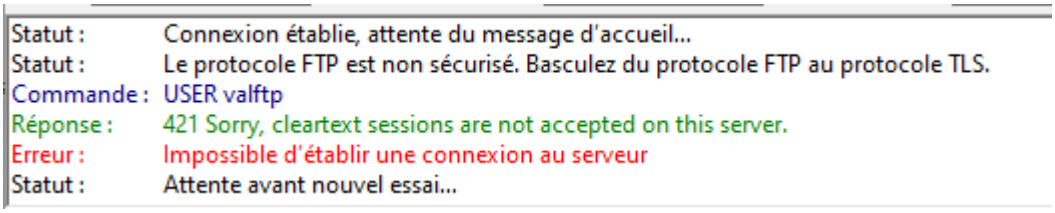


Schéma de la réalisation du TP :

